

CHAPTER 6 - FRONTEND DEVELOPMENT

6.1 Introduction

The frontend of EduGenie provides a simple and interactive interface for students to access AI-powered educational services. It was developed using HTML and CSS and communicates with the FastAPI backend.

6.2 Frontend Technologies

HTML5 is used for page structure, CSS3 for styling, and Jinja2 templates for rendering pages. These technologies provide a lightweight and responsive interface.

6.3 User Interface Design

The interface contains sections for Question Answering, Explanation, Quiz Generation, Summarization, and Learning Path Recommendation. Each section accepts user input and displays AI-generated output.

6.4 Frontend Workflow

The user opens the application, enters a query, submits the request, and receives the generated response after backend processing.

6.5 Frontend–Backend Integration

The frontend sends HTTP requests to FastAPI endpoints. FastAPI processes the requests using Google Gemini or LaMini-Flan-T5 and returns the results to the browser.

6.6 User Experience

The interface is easy to navigate, responsive, and designed to reduce the learning curve for students. The layout is simple and focuses on usability.

6.7 Advantages

Simple interface, responsive design, fast interaction, easy backend integration, modular architecture, and improved user experience.

6.8 Chapter Summary

This chapter described the frontend implementation, technologies used, interface design, workflow, integration with the backend, and overall user experience of the EduGenie application.